

A disproof of conjecture 00000000982: fixed points of the Berezin transform on Fock space

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Abstract

Conjecture 00000000982 asserts that the fixed points of the Berezin transform on Fock space are radial functions and that the fixed-point set has cardinality at most 2. We show that the conjecture is **false** under its literal reading. On the Fock space with the Gaussian measure $\frac{1}{\pi}e^{-|w|^2}dA(w)$, the (diagonal) Berezin transform is the Gaussian convolution

$$Bf(z) = \frac{1}{\pi} \int_{\mathbb{C}} f(w) e^{-|w-z|^2} dA(w) = e^{\Delta/4} f(z),$$

so it fixes every harmonic function and therefore every element of the holomorphic Fock space F^2 . The function $f(z) = z$ belongs to F^2 (it has $\|z\|^2 = 1$) and is fixed, but it is not radial: $|1| = |i| = 1$ while $z(1) = 1 \neq i = z(i)$. Moreover $1, z, z^2, \dots$ are pairwise distinct fixed points, so the fixed-point set is infinite and its cardinality is not at most 2. The strongest possible objection is that “on Fock space” might be intended to mean “on the bounded symbols” (the Toeplitz algebra); under that reading Liouville’s theorem makes the fixed set the constants — radial of cardinality 1 — and the conjecture would be a garbled statement of a true fact. Under the literal reading, however, it is false as exhibited.

1 The Fock space and the Berezin transform

Definition 1 (Fock space). Let dA denote Lebesgue area measure on $\mathbb{C} \cong \mathbb{R}^2$. The *Fock space* is

$$F^2 = \left\{ f : \mathbb{C} \rightarrow \mathbb{C} \text{ holomorphic} : \|f\|^2 = \frac{1}{\pi} \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} dA(z) < \infty \right\}.$$

It is a Hilbert space with inner product $\langle f, g \rangle = \frac{1}{\pi} \int_{\mathbb{C}} f(z) \overline{g(z)} e^{-|z|^2} dA(z)$. The monomials $\{z^k\}_{k \geq 0}$ are orthogonal with $\|z^k\|^2 = k!$; in particular $\|1\|^2 = 1$ and $\|z\|^2 = 1$.

Definition 2 (Berezin transform). For a function f for which the integral converges, the (*diagonal*) *Berezin transform* is

$$Bf(z) = \frac{1}{\pi} \int_{\mathbb{C}} f(w) e^{-|w-z|^2} dA(w).$$

This is the average of f against the normalized Gaussian (the diagonal coherent-state weight) centred at z . Writing $w = z + u$ and using translation invariance of dA gives the convolution form

$$Bf(z) = \frac{1}{\pi} \int_{\mathbb{C}} f(z + u) e^{-|u|^2} dA(u).$$

Remark 3. All of F^2 consists of entire functions of order ≤ 2 ; for such f the Gaussian integral in Definition 2 converges absolutely for every z and the manipulations below are justified.

2 The convolution identity $B = e^{\Delta/4}$

Write $\partial = \partial_z$, $\bar{\partial} = \partial_{\bar{z}}$ for the Wirtinger derivatives and $\Delta = 4\partial\bar{\partial}$ for the Laplacian.

Lemma 4 (Gaussian moments). *For integers $m, n \geq 0$,*

$$\frac{1}{\pi} \int_{\mathbb{C}} u^m \bar{u}^n e^{-|u|^2} dA(u) = m! \delta_{mn}.$$

Proof. In polar coordinates $u = re^{i\theta}$, so $dA = r dr d\theta$ and $u^m \bar{u}^n = r^{m+n} e^{i(m-n)\theta}$. The angular integral is $2\pi\delta_{mn}$, and for $m = n$, $\frac{1}{\pi} \cdot 2\pi \int_0^\infty r^{2m+1} e^{-r^2} dr = 2 \cdot \frac{1}{2} \Gamma(m+1) = m!$. $\square$

Theorem 5 (Convolution identity). *For every entire f of at most exponential growth of order 2 (in particular for every $f \in F^2$),*

$$Bf(z) = \sum_{m \geq 0} \frac{(\partial^m \bar{\partial}^m f)(z)}{m!} = e^{\partial\bar{\partial}} f(z) = e^{\Delta/4} f(z),$$

the series converging absolutely.

Proof. Expand $f(z+u)$ in its Taylor series at z :

$$f(z+u) = \sum_{m,n \geq 0} \frac{(\partial^m \bar{\partial}^n f)(z)}{m! n!} u^m \bar{u}^n.$$

Integrating term by term against $\frac{1}{\pi} e^{-|u|^2} dA(u)$ and applying Lemma 4 kills every term with $m \neq n$ and leaves

$$Bf(z) = \sum_{m \geq 0} \frac{(\partial^m \bar{\partial}^m f)(z)}{m! m!} \cdot m! = \sum_{m \geq 0} \frac{(\partial\bar{\partial})^m f(z)}{m!} = e^{\partial\bar{\partial}} f(z).$$

Since $\partial\bar{\partial} = \Delta/4$, this is $e^{\Delta/4} f(z)$. Absolute convergence on compact sets follows from the order-2 growth of f and the Gaussian decay of the kernel. $\square$

Corollary 6 (Harmonic kernel). *On the space of polynomially bounded C^∞ functions,*

$$Bf = f \iff \Delta f = 0,$$

i.e. the fixed points of B are exactly the harmonic functions. In particular every holomorphic function is fixed.

Proof. If $\Delta f = 0$ then $\partial\bar{\partial}f = 0$ and Theorem 5 gives $Bf = f$ immediately; holomorphic f satisfies $\bar{\partial}f = 0$ and hence $\Delta f = 4\partial\bar{\partial}f = 0$.

Conversely, $B - 1 = e^{\Delta/4} - 1 = (\Delta/4)\psi(\Delta/4)$ where $\psi(t) = (e^t - 1)/t = \sum_{k \geq 0} t^k/(k+1)!$ is entire with $\psi(0) = 1$. On the Fourier side $e^{\Delta/4}$ multiplies $\hat{f}(\xi)$ by $e^{-|\xi|^2/4}$, so $Bf = f$ forces $(e^{-|\xi|^2/4} - 1)\hat{f}(\xi) = 0$ for all ξ . The first factor vanishes only at $\xi = 0$, so $\hat{f}$ is supported at the origin and f is a polynomial. On the finite-dimensional space of polynomials of degree $\leq N$ the operator $\psi(\Delta/4)$ is invertible (it is a polynomial in Δ with constant term 1, triangular with respect to the degree filtration), so $(\Delta/4)\psi(\Delta/4)f = 0$ implies $\Delta f = 0$. Hence $Bf = f$ implies $\Delta f = 0$. $\square$

Remark 7. Harmonic functions need not be holomorphic: $z + \bar{z} = 2\operatorname{Re} z$ is harmonic but not holomorphic. Such functions are fixed by B as elements of the ambient space $L^2(\mathbb{C}, \frac{1}{\pi} e^{-|w|^2} dA)$, but they are *not* elements of the holomorphic Fock space F^2 . The correct inside- F^2 witness for the refutation is $f(z) = z$ (see below).

3 The fixed point z is not radial

Definition 8 (Radial). A function g on $\mathbb{C}$ is *radial* when it depends only on $|z|$, i.e. $g(z) = g(w)$ whenever $|z| = |w|$.

Theorem 9 (Non-radial fixed point). *The function $f(z) = z$ belongs to F^2 , satisfies $\|f\|^2 = 1$, is a fixed point of B , and is not radial. Consequently the fixed-point set of B is not contained in the radial functions, and the first half of conjecture 00000000982 is false.*

Proof. That $z \in F^2$ with $\|z\|^2 = 1$ is the monomial computation of the Definition of F^2 . It is holomorphic, so by Corollary 6 (or directly, since $\bar{\partial}z = 0$) $Bz = z$.

It is not radial: the points 1 and i have equal modulus $|1| = |i| = 1$, but

$$z(1) = 1 \neq i = z(i).$$

Thus z takes different values on a common modulus level set, so it is not radial, yet it is fixed. $\square$

Remark 10. Neither $\operatorname{Re} z$ nor $\operatorname{Im} z$ may be used here: they lie in the ambient $L^2(\mathbb{C}, \frac{1}{\pi} e^{-|w|^2} dA)$ but not in the holomorphic Fock space F^2 . The in- F^2 witness is $f(z) = z$.

4 The fixed-point set is infinite

Theorem 11 (Infinite fixed set). *For every $k \geq 0$ the monomial z^k is a fixed point of B ; the monomials $\{z^k\}_{k \geq 0}$ are pairwise distinct. Hence*

$$F^2 \subseteq \operatorname{Fix}(B), \quad \operatorname{Fix}(B) \text{ is infinite,}$$

and in particular $1, z, z^2$ are three pairwise distinct fixed points. The cardinality of the fixed-point set is therefore not at most 2, so the second half of conjecture 00000000982 is false as well.

Proof. Each z^k is holomorphic and hence, by Corollary 6, fixed: $Bz^k = z^k$ (directly, $\bar{\partial}z^k = 0$ for $k \geq 0$). The monomials are distinct as functions: $z^k \neq z^\ell$ for $k \neq \ell$ (evaluate at any nonzero point and compare degrees, or note the values at $z = 2$ are $2^k \neq 2^\ell$ for $k \neq \ell$). For instance

$$1 \neq z, \quad 1 \neq z^2, \quad z \neq z^2,$$

where the last two are distinguished at $z = 0$ and $z = 2$ respectively. Thus $\operatorname{Fix}(B)$ contains the infinite linearly independent set $\{z^k\}$, so it is infinite, and it contains at least the three distinct points $1, z, z^2$. $\square$

5 The bounded-symbol objection

The strongest objection to the above is a change of reading. If “on Fock space” is taken to mean “on the algebra of bounded symbols” — the Toeplitz algebra of F^2 , acting on F^2 — then the Berezin transform still acts as $e^{\Delta/4}$ on symbols, but now one restricts to *bounded* symbols. A bounded harmonic function on $\mathbb{C}$ is constant by Liouville’s theorem: if $\Delta f = 0$ and f is bounded, then f is constant. Corollary 6 then gives

$$\operatorname{Fix}(B) \cap L^\infty(\mathbb{C}) = \{\text{constants}\},$$

which is radial and has cardinality exactly 1. Under that reading the conjecture “fixed points are radial, and there are at most 2 of them” is a garbled statement of a true fact (the cardinality bound should be 1, not 2, and one must restrict to bounded symbols).

We record this honestly, but it does not rescue the conjecture as written. The statement in the conjecture file is about fixed points “on Fock space”, i.e. elements of F^2 ; there the fixed set is all of F^2 , is far from radial, and is infinite. The literal reading is false.

6 Summary of the refutation

1. **Convolution identity.** On the Fock space with Gaussian measure, $Bf(z) = \frac{1}{\pi} \int f(z + u) e^{-|u|^2} dA(u) = e^{\Delta/4} f(z)$ (Theorem 5).
2. **Harmonic fixed points.** $Bf = f$ iff f is harmonic, so every holomorphic $f \in F^2$ is fixed (Corollary 6).
3. **Non-radial witness.** $z \in F^2$ is fixed but not radial: $|1| = |i|$ while $z(1) \neq z(i)$ (Theorem 9). The first claim of the conjecture fails.
4. **Infinite fixed set.** Every z^k is fixed and the z^k are pairwise distinct, so $\text{Fix}(B)$ is infinite; in particular its cardinality is not ≤ 2 (Theorem 11). The second claim fails.

Numerical confirmation. The script `reproduce.py` evaluates B by tensor Gauss–Hermite quadrature (40 nodes per dimension, exact for polynomials of degree ≤ 79) on a grid of complex points. It confirms, to about 10^{-15} ,

$$B(1) = 1, \quad B(z) = z, \quad B(z^2) = z^2, \quad B(z^3) = z^3,$$

together with $B(z + \bar{z}) = z + \bar{z}$ (a harmonic, non-holomorphic fixed point) and the negative control $B(|z|^2) = |z|^2 + 1$ (non-harmonic, not fixed). It also verifies the non-radiality witness $|1| = |i|$ with $z(1) \neq z(i)$. The accompanying core-Lean development `lean4/` machine-checks the distinguishing finite facts: that z is not radial, and that $1, z, z^2$ are pairwise distinct, over the integer lattice $\mathbb{Z} \times \mathbb{Z} \subset \mathbb{C}$; the analytic identity $B = e^{\Delta/4}$ and the harmonic-kernel characterisation are proved here and checked numerically.

Verdict. Conjecture 00000000982 is **false** under its literal reading: the fixed points are not radial and are infinite in number. Under a bounded-symbol reading it reduces to a garbled version of the true statement that the only bounded harmonic fixed points are the constants.